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1 Introduction

The power to analyze relationships between large amounts of data has never been as accessible as it has been with the development of machine learning algorithms. However, there is only so much data in the world so companies that want to push what their products can do rely on large unmonitored sets of data. This can cause algorithms to be biased towards minority groups. This issue exponentiates when the usage of this technology now stretches from healthcare, recruitment, law enforcement and more. This essay will attempt to focus on salient issues brought up by the 2020 documentary "Coded Bias".

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2 Background information

The machine learning market has been developing rapidly since 2020, with the market value being projected to be around 282 billion dollars in size. Because of this rapid development, there is a larger focus on the usage of these models and the harm they can cause.

In 2020, the documentary, "Coded Bias" was released on Netflix. This documentary focuses on the research of MIT media researcher Joy Buolamwini. Her research focuses on the data collection issue plaguing machine learning models' ability to not discriminate. This research started because facial recognition algorithms were found to have significant difficulties in identifying faces of people of color and women. [Kantayya 2020]

The documentary not only covers Buolamwini's research but also the impacts of machine learning models. One notorious example is the group Big Brother Watch in the UK which studies how the biased facial recognition systems trigger law enforcement systems with false positives and cause major stress for individuals. [Big-Brother-Watch 2020]

Lastly, the documentary explores how there could be possible improvements for both the algorithms and the structures that produce the models. There is a heavy emphasis on the lack of active regulation and how the law needs to keep up with the technology being developed to prevent further issues from arising. [Sikar 2025]

2.1 Stakeholders

There are several stakeholders in this scenario with a major power imbalance between the machine learning companies and the general public:

- Machine Learning Companies: As they decide what gets made and how it is made
- Machine Learning Engineers: As they implement the algorithms and methods of training
- Governments: As they regulate and use the algorithms for various purposes
- Software companies: As they use the algorithms to enhance their businesses
- General public: As they are the ones who are being judged by the algorithm

This power imbalance places the general public at a major disadvantage as machine learning algorithms rely on pre-existing data which is not guaranteed to be sanitized of biases. Therefore causing unjust profiling of individuals, rejections in jobs because of nationality/ethnicity, and more without any form of control as the mathematical models are a black box and companies have no incentive to reverse-engineer it.

3 Salient Issues

The documentary's main argument is that in computer systems that rely on machine learning are not accounted for in the USA's legal system at a reasonable pace. Therefore allowing companies to neglect the possible biases that the algorithms might have. This can and has been causing violations of human rights as different industries adopt this technology to handle decision-making.[Big-Brother-Watch 2020]

Since the original studies and the documentary, the usage of machine learning has increased across all industries [Rashid and Kausik 2024]. This in turn further pushes the need for boundaries in the development, training and usage of said models by major entities like multi-national companies and governments across the world. In this essay, there will be an attempt to apply ethical frameworks to evaluate the following questions:

- What moral duties do ML companies have when producing their models?
- Who would be acting unethically for not taking responsibility when the models cause damage?
- Is it ethically feasible to enforce regulations in the black box that are ML model?

4 Technological Background

To be able to evaluate the biases of machine learning models, the production of said models must first be understood to derive the links between the stakeholders and the impacts of the algorithms.

4.1 Machine Learning Types

Machine learning is a subset of artificial intelligence that uses statistical inference to adapt responses based on previous and new knowledge. This is a very powerful type of program as it provides adaptability at no code architecture change, however, these systems improve drastically the more data they have available and have analyzed. To achieve this functionality a ML model must first be trained. There are 5 key types of training that can be done on a model [China NA]:

4.1.1 Supervised Learning. This type of training is done with a set of labelled data so the model can build a relationship between the provided labels and the data. This type of algorithm is heavily used in image recognition, risk assessment and fraud detection.

4.1.2 Unsupervised Learning. Unlike the previous method, this method tries to find unique labels to learn around. This is used in predictive modelling and pattern recognition.

4.1.3 Self-supervised Learning. Like unsupervised learning, this method doesn't require the labels. The main feature of this method is that it will learn the input from elsewhere and turn unsupervised problems to supervised problems by generating labels. This is common in computer vision and natural language processing.

4.1.4 Reinforcement Learning. This method is unlike the previous ones where there is a data set, in this method the response of the algorithm's output is rated as good or bad and through iteration it will

approach the desired behaviour. This method is mainly used to train robots and game behaviours.

4.1.5 Semi-supervised Learning. This method is a mix of the 2 first methods. It is commonly used in most contexts to find relationships not labelled at the beginning of training.

4.2 Data Collection and Cleaning (Add more sources)

The process of data collection is a key process in the development of any model, it is the core of the algorithm's ability to adapt. Therefore the data collection pipeline is complex to reduce undesired errors. The pipeline is complex and composed of many phases [Vidhya et al. Jun].

4.2.1 Data Collection. This processes involve obtaining and labelling the data. They have the main limitation of cost, providing human made labels for models is expensive. There are also issues with ensuring that the data sources are of quality and large volume which is what data acquisition and discovery do. This part of the development of models is essential to ensure the model is trained on data which is not poisoned or biased.

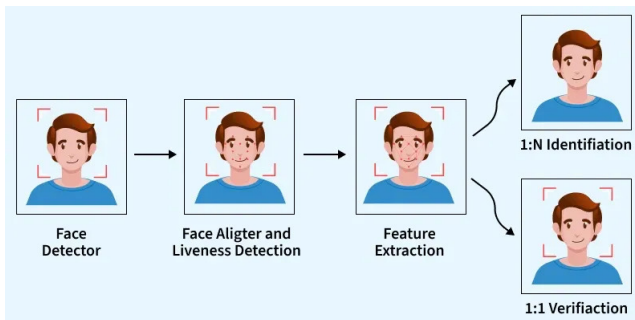
4.2.2 Data Cleaning. After the data is trained, the data needs to be analyzed for possible errors. This is the stage where the data can be validated through humans and other algorithms to ensure that the data provides results which are not noisy. This is a key part of ensuring that a model can be fair by checking the following:

- Majority groups are over-represented
- Data for minority groups is poisoned

4.2.3 Fairness Training. To solve this Distributional Robust Training is done which is an iterative process where the weights are adjusted to converge at impartiality. There is also FR training which is a comparison process with a clean model and a trained model which is meant to protect and also remove the biases of the model without sacrificing the accuracy.

4.3 Facial Recognition (Add sources)

Facial recognition is a computer vision algorithm that is used to identify individuals against a database of stored faces for various purposes ranging from security to healthcare. Modern facial recognition is powered by machine learning as it increases the speed and effectiveness of the model when dealing with non-ideal conditions.



FigureFacial recognition pipeline (SOURCE FROM GEEKS4GEEKS HERE)

4.3.1 Face Detection and Alignment. Face detection is the process of detecting a face is in the frame, this can be implemented with and without machine learning but modern methods get more speed by using it in the process. After a face is identified, it needs to be framed to run the next parts of the algorithm, this uses facial features to locate key details in non-ideal conditions.

4.3.2 Feature Extraction. For this stage, the scanned features are converted into a numerical vector which is then used with machine learning to calculate similarity score to be able to locate faces in the database.

4.3.3 Face Matching. From the obtained vector, the values are then verified by measuring the different facial features through geometrical analysis and machine learning models for comparison.

5 Ethical Analysis

With the context required, it is now possible to evaluate the salient issues presented by the documentary. While all questions will be evaluates using workable ethical frameworks, they will be different due to the compatibility between the question and the framework's ability to judge it without devolving to relativism.

5.1 Responsibilities of ML companies

When an ML company decides to produce a model, what are the responsiblities do they have towards all the involved stake-holders? The lack of governance over ML implementations has brought up the need to discuss what are the expectations towards a company's model.

5.1.1 Kantianism. Under Kantianism, companies are presented with the following formulations when it comes to a model's bias:

(1) If a user damages someone with my product, it was a user's error
(2) Lying to others is ethically sound if I don't know better
(3) Making money off models that misrepresent people is ethical

For these formulations it can be seen that there are major contradictions. If everyone could blame their clients for the mistakes that their products make there would be no trust in any industry and therefore no development. For the second formulation runs into a contradiction, where if everyone were allow to spread misinformation, there would be no source of truth for humanity. Lastly, the 3rd formulation is removing humanity out of the minorities in the situation by neglecting the pain that the misrepresentation can cause. Evidence of this pain can be seen in the documentary as people of color are persecuted unfairly which causes them great distress and removes their agency when being judged. (CITE SOURCES HERE)

5.1.2 Social Contract Theory. Under social contract theory, instead of evaluating imperatives, there are implicit expectations that we as a society agree upon when integrating a system where machine learning is used:

(1) The algorithm will measure biases impartially as it has no previous lived experiences
(2) The companies are developing the product with the goal of betterment of society with financial gain as compensation (as is the case with most development under capitalism)

As seen in the technical section, ML models do not poses the ability to be impartial as they reflect the biases of the sources without major human intervention which does not reflect the current usage of the tool. (ADD SOURCES HERE)

5.1.3 Act Utilitarianism. Compared to the other frameworks, the answer to our question can be evaluated via the maximization of the collective joy of people. Under this, it is hard to account for all possible factors but there is a majority which would benefit from the reduced cost of the models. However, discrimination brings issues to a population at large regardless of their level of privilege (SOURCE HERE). This helps reframe the possible measurements of joy from members of the majority to members of the minority to a measurement of how much the joy of an entire population weighs against those benefiting from the harm being caused by implementing models without strict regulation and cost. (ADD SOURCE HERE)

In 2025, it was reported that Immigration Customs Enforcement (ICE) in the USA has been utilizing a mobile app for identifying undocumented immigrants. That same year there had been several cases where citizens and documented immigrants were being unlawfully deported. On the other hand, using machine learning for law enforcement can greatly enhance safety when used correctly such as processing body camera footage from cops, predict threats via behavioral patterns and provide support to those who need it. Therefore, this makes the value society can get from it heavily vary in the fairness of the models used

making it unethical with the current state of the technology. (ADD SOURCES HERE)

5.2 Stakeholders responsible for damage

When approaching the issue of creating legislation for ML models, there is a need to focus on the damages and how ethical frameworks can represent the damage being done:

5.2.1 Social Contract Theory. Under social contract theory, there is the ideal of assigning conditions before getting a role but in practice it functions the opposite. This causes an issue to arise when dealing with the responsibilities as a veil of ignorance can't really be kept as the models are already owned and managed by a select few. Because of this, it is important to focus on the pre-existing contracts established by society such as discrimination being harmful and harm being bad.

In this case, the main difference from an individual discriminating another is the discrimination is a sequence of processes. First, the data is collected from sources that carry bias, then the models are trained on this biased data and sometimes iterated upon to reduce bias. From there the ML company sells the model to a client without clearly labelling the bias in the model then a client proceeds to use it. As there are 2 beneficiaries of this technology, in social contract theory it is part of the contract that lying to clients is unethical, both from the ML company and the client company cases. From this, it is possible to determine that the ML company is complicit in the damage caused by the client's usage as they may not provide transparency in the model's bias. On the other hand, the client company is responsible for the damage they caused for not verifying their products would not harm their customers. This is because under social contract, people agree that lying and hurting your customer base is unethical.

5.2.2 Virtue Ethics. In virtue ethics, the assessment of who is responsible can be seen through the virtues of honesty and fairness. With this said, it can be seen that if machine learning algorithms discriminate in the hiring process, it is doing an unfair judgement, the person responsible for that judgement would be the person who decided to use the machine's verdict to determine such a thing. This shows that they were not practicing the virtues of fairness as they trusted a weighted model. In the case the model was not labelled as biased the party to blame is the development team for their lack of honesty in their sale of a model.

5.3 Ethical feasibility of enforcing regulations

As discussed in the previous sections, there are several issues that arise with the usage of machine learning to judge individuals, however, it is important to evaluate whether these processes can be purified to be ethical. This is a major issue as providing a product free of bias would mean it needs a lot of data refinement without guarantee that it will be rid of biases.

5.3.1 Kantianism. Kantianism is based on principles of universality, this means that if a formulation can be applied universally without contradiction, it is ethical. Therefore, the formulation "a machine is allowed to judge an individual unfairly" would devolve into the issue where machines could be used for increasingly more harmful things such as judging whether someone lives or not by having access to a job, healthcare or criminal punishment. If this were the case it would create a situation where machines are used to disregard other individual's humanity. Therefore making it unethical under Kantianism unless the model is proved to be fair, which is impossible with the current technologies.

5.3.2 Act Utilitarianism. Unlike the previous framework, the usage of a model can be ethical if the benefits outweigh the negatives. As discussed in the responsibilities of stakeholders, there is a very big emphasis on whether the model can produce numerical gains. This provides a stronger argument for using models in surveillance as they will

keep the majority safe while the people who get affected are not as accounted for. The bigger issue is that just like mentioned in the previous categories, discrimination hurts the group in the long term as intra-group conflict causes more damage in terms of happiness and sustainability. Therefore, the ethical aspect of these models provides a larger loss in the case that it discriminates to a substantial extent. On the other hand, if these systems work a reasonable amount, it could be considered ethically sound.

6 Conclusion

To conclude, machine learning models pose several benefits in terms of technology and the ability to make infinitely adaptable algorithms. Just like any powerful tool, it must be managed for good, if the goals of machine learning companies is simply to make money, this is not guaranteed to bring betterment to society. Therefore, it is important to ensure there is regulation and incentive for machine learning developers, researchers and clients to seek to improve their practices. By looking at the ethical frameworks it was possible to derive that there isn't a single stakeholder to blame but it is more of a collective effort between the general public, the government and the companies that use/sell machine learning systems. In the documentary, there is a scene where they mention that this is a continuous effort and that regulation is not a one and done, which is reflected by the last question asked by this paper, these models reflect the bias in society therefore to use them more ethically the bias must be either acknowledged or the models must remain unused at all. On the more optimistic side, there is a lot of research into optimizing the fairness of models as removing bias has a lot of incentives beyond removing the biases towards people. The commercial incentive might drive the technology to develop to the point models meet the ethical requirements for a better world.

7 Beyond the superficial

Machine learning is a technology that is here to stay, as software engineers, it is important to understand the social impacts of the technology we develop. Given the world is now more interconnected as well as polarized, there is a high importance in the way that technology affects all possible types of customers. In the case of DigiPen students, there are a lot of game developers that work with peripherals that might be using machine learning, understanding how the technology affects your players is important to ensure that they can have a good time and the art being made is not solely for a privileged majority.

On the case of other DigiPen students, there are some that go into embedded/imaging systems. Given the growth of this technology towards self-driving vehicles and robotics, it is very important as an engineer to try and acknowledge the ways in which your algorithms weigh different individuals involved. Be it from a medical device's ability to properly find patterns across a wide range of patients to a robotic arm at a factory being able to properly identify the other workers to ensure safety.

At a personal level, the greatest take away is that technology moves really fast compared to all the available support systems. This is scary but the fear is not for silence, it's more of an encouragement for me to make the world a better place through my work while being conscious of what I bring to the table. As discussed in the first paper for this class, I believe that reliable software is a requirement for software to bring value to society. This belief extends to this, for a system to be reliable it must provide answers at a deterministic level or substantially so. The usage of machine learning is not deterministic but also provides several benefits to society but a reliable system knows exactly what issues it has and how it can be overcome either through itself or other external systems.

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